

MORE BOARD GAME EDUCATION

Inspiring Students Through Board Games

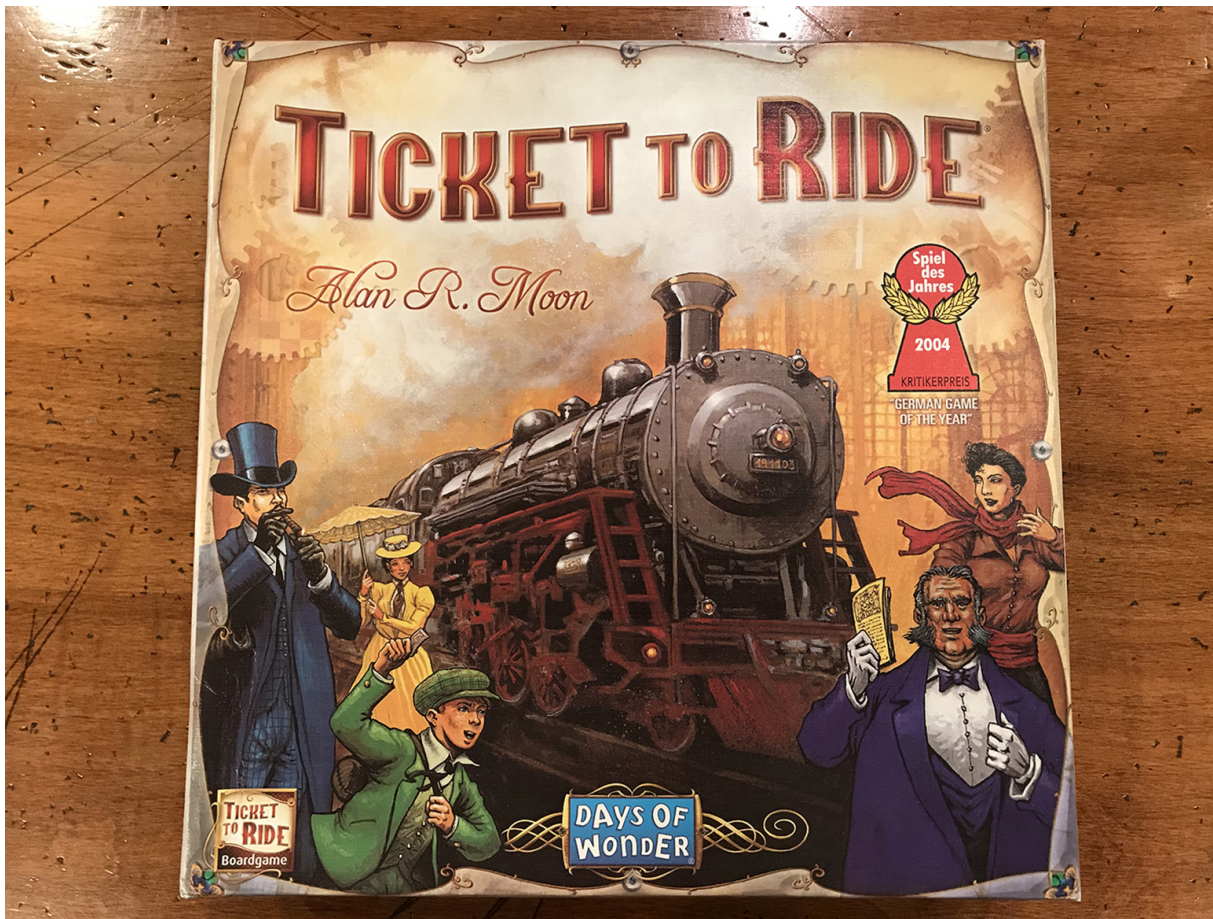
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Chapter 3

Ticket to Ride

All Aboard . . . For Learning!





Who doesn't love trains? More to the point, who doesn't love the idea of constructing rail lines to destinations across countries in every part of the world, while racing to complete your rail line and trains before your opponents? In our family our kids were obsessed growing up with all things "*Thomas*" and my wife and I spent countless hours on the floor or on our homemade train table connecting wood tracks with "cheeky little engines" and Thomas, Edward, Toby, and the whole crew. (Including a castoff Lego "Penguin" character who filled in very nicely as Sir Topham Hatt.)

As such, when we first played Ticket to Ride it became an instant favorite with our whole family. Ticket to Ride involves everything people love about trains; but does so through a game that not only develops problem-solving, reasoning, and critical thinking skills, but also educates in the areas of geography and engineering: two skill sets not commonly engaged in board games. All aboard!

A BRIEF HISTORY AND DESCRIPTION OF TICKET TO RIDE

Released in 2004 by Days of Wonder, Ticket to Ride was invented by Alan R. Moon. Inspired by his childhood memories of playing Monopoly and Risk, Mr. Moon can best be described as a board game junkie and he worked for decades designing board games while he waited tables.

After numerous prototypes and failed attempts, Mr. Moon eventually came up with Ticket to Ride and presented the game to three publishers. Of those publishers, only Days of Wonder responded. (The other publishers probably have some understanding of how the Portland Trail Blazers felt when they chose not to draft Michael Jordan in 1984.) Ticket to Ride went on to win the coveted Spiel des Jahres (Game of the Year) in 2004.

Interestingly, while the initial sales of Ticket to Ride were solid, sales really took off when Days of Wonder made the game available on line for free. While seemingly counter intuitive, introducing this "German style" game to millions of people generated incredible interest in the board game version. To date, over four million copies have been sold with sales in excess of \$200 million. Mr. Moon no longer waits tables.

While the original version was set in North America, multiple subsequent versions of Ticket to Ride cover Europe, Asia, Africa and Scandinavia. Moreover, the subsequent versions add additional features and game play which will briefly be touched on.

Common to all versions, Ticket to Ride is played by players attempting to claim and complete routes to specific destinations which they have drawn. Points are awarded for completing routes based on their length and additional points are awarded if the route connects destinations a player has drawn. However, destinations are kept secret and other players can also claim these routes, which results in negative points for a player who does not complete routes to her drawn destinations. Bonus points are also awarded for the longest continuous route completed.

Subsequent versions of Ticket to Ride added additional features and opportunities to collect points, but the basic scoring and game play is consistent across all variations. As an example, the Europe version introduced routes requiring players to obtain Ferry and Tunnel cards. The Marklin versions provide the opportunity to earn additional points for merchandise tokens and can be played with a dice option. This, in part, is what makes Ticket to Ride so engaging in that it is simple enough for a younger player to understand but also allows for more sophisticated and complex game play in some of the subsequent versions.

SKILLS DEVELOPED

Ticket to Ride develops and hones a plethora of core educational skills, including math, problem-solving, creativity, and deductive reasoning. However, the skill set that sets this game apart from almost all other games is its focus on geography and history. Indeed, the very nature of the game immerses players in the geography of countries and regions all over the world.

Since the board is a map, Ticket to Ride requires players to develop spatial and directional skills as well as map reading skills. Moreover, the cities used in Ticket to Ride are real world locations which are not only simply places on the map, but act as starting points and destinations and require players to comprehend the distances and relative locations of those cities. Understanding that Little Rock is south of Helena, and a long distance apart, is knowledge that many students simply will not retain by reading it in a book.

The geography skills taught through Ticket to Ride also present incredible history and economic lessons. How so, you might ask? The cities highlighted in Ticket to Ride include locations that are not generally familiar to current generations as the game is set at the turn of the twentieth century. Places like Saul St. Marie, Duluth, Little Rock, and Raleigh are certainly not as familiar to most players as New York, Chicago, and Los Angeles. This provides an excellent opportunity for players to explore why some of these lesser known cities were considered important destinations for railroads. Moreover,

the game provides a visual depiction of how cities like Duluth and Saul St. Marie, with both rail and deep-water ports and in close proximity to iron ore and other raw materials, were critical to the industrial development of the country.

Like many of the games discussed in this book, Ticket to Ride excels in developing reasoning and critical thinking skills. Players need to evaluate multiple options to determine which strategy will provide the most points. Some games that may mean claiming as many routes as possible versus working toward completing the longest route or destination routes. However, these strategic decisions cannot be made in a vacuum as each player must also analyze and deduce her opponent's strategy to identify the critical and contested routes to allow for higher scores. Ticket to Ride requires each player to work through these strategy questions on each move, and to do so with an understanding of the number of turns left in the game.

Along the same lines, deductive reasoning plays a significant role in playing Ticket to Ride. By observing, and keeping track of, the color of the train cards selected by a player's opponents a player will be able identify routes available to that player. Understanding the routes available to a player's opponents, in conjunction with the routes they select, also allows a player to deduce the destination strategies of his opponents. Of course, blind draws of train cards reduce the ability of a player to track color selections but over the course of multiple games players will develop deductive reasoning skills that will allow them to identify opponents' strategies with less than perfect information.

Ticket to Ride also fosters development of critical math skills, especially for younger players. As all routes have different point values, players are required to compare and contrast different route alternatives with their corresponding point values. Not only are players constantly required to calculate point values available for each move, they also must weigh their moves against the point values that are also available to their opponents. All these calculations must be made with the backdrop of the available pieces for each player since a player cannot obviously complete two 6 segment routes if he only has ten pieces left. The best part of all of this is that since these

multiple calculations are simply part of the game players do not even realize the extensive math involved!

One of the unique features of Ticket to Ride is that there are multiple options between destinations, especially those destinations in different parts of the country. For example, a direct route from Pittsburgh to Denver through Chicago and Omaha can be accomplished with eleven train pieces. However, while this route will result in eighteen points a route from Pittsburgh to Denver through Saint Louis and Kansas City, also using eleven train pieces, will garner nineteen points. Players will necessarily engage in multiple mental math calculations planning strategy in determining each move. Such mental math calculations will generally involve the addition of multiple numbers and the comparisons of the sums with other multiple number calculations: the same type of exercise that draws groans in math class.

A final component of the math skills developed by Ticket to Ride is the game's use of a number line for scoring. The use of a number line teaches ordering as well as addition skills. With the possibility of uncompleted destination cards in their possession players in Ticket to Ride will also develop subtraction and netting out skills, and the use of the number line provides an excellent visual tool to help younger players develop these critical base math skills.

The numerous available routes on the Ticket to Ride game board, as well as the multiple ways in which players may earn points, develop another critical educational skill: creativity. While initial observation may lead a player to believe that completing direct routes between destinations is the way to score the most points, Ticket to Ride's multiple scoring opportunities soon teaches otherwise.

With respect to destination cards, while direct routes may initially appear to provide the most scoring opportunities, they also lead to obvious strategies that can be blocked by opponents. Rather, players learn to look at routes to destinations creatively and, by factoring in points for the longest route and multiple destinations, discover that longer and indirect routes may turn out to be a more successful strategy. However, because there are multiple scoring options and opponents change strategies from game to game players learn to

evaluate alternative moves and strategies as the games progress. Since there is no perfect strategy for every Ticket to Ride game players are free to explore and experiment with multiple strategies and options thereby developing creativity and flexibility, skills that are inherently necessary to problem-solving in any context.

One of the other characteristics of Ticket to Ride that makes it such an appealing game is the ease by which younger players can compete on a level playing field. While the game suggests ages of eight and above, the game is easily learned by players as young as four and five. Color pattern recognition and grouping skills are bonus core educational concepts that younger players will also develop through Ticket to Ride. Indeed, the act of placing the train pieces on the designated spaces on the board when completing routes even develops manual dexterity for both younger players and players who may have developmental challenges.

VARIATIONS

The unique set up of Ticket to Ride makes it one of the best games for variations and tweaks to focus on particular skill sets. In particular, because of the focus on maps Ticket to Ride provides unlimited opportunities to hone geography skills.

As a starting point, the numerous official editions of Ticket to Ride focusing on Germany, the Nordic countries, Europe, Asia, and Africa offer opportunities for players to become familiar with the geography of a variety of countries and regions across the globe. However, to take the geography lessons a step further use a large continent, country or even state map to create your own Ticket to Ride game board. Creating a customized game board using a map with real destinations allow a focus on any specific geographic area. Moreover, this is an excellent way to develop a geographic understanding of regions of the world which are generally not well known but in which current events are taking place. For example, what a great way to understand the geographic relationship of Syria, Pakistan, Iran, and Israel!

An additional tweak on Ticket to Ride is to require players to identify the state or countries across which routes traverse before they can claim the route. Since the game board does not identify the states or countries players will need to learn the states and countries, which will also develop an understanding of where various states and countries are located. Believe it or not, a few games of Ticket to Ride employing this additional rule will result in players being able to identify all the continental states and most countries: a much better alternative than trying to memorize a state map.

Another variation that we used extensively with our kids and in the classroom is to use state maps. Most schools in the United States dedicate a year to learning state history and geography. What better way to learn state geography and all those annoying county seats than by using the state map to play Ticket to Ride!

As mentioned previously, one of the unique features of Ticket to Ride is that it uses real city destinations. This feature also provides a fantastic opportunity to incorporate history and historical facts into the game play. By providing additional points for each historical fact that a player can identify with respect to connected cities after completing a route, Ticket to Ride becomes the perfect tool to reinforce and teach history. For example, awarding a point to a player who identifies that Duluth was noted for its iron ore supply develops a historical understanding of the destination cities that goes well beyond simply connecting orange and red train pieces.

Along these same lines, because Ticket to Ride focuses on transportation and the railroads at the dawn of the twentieth century it also provides an excellent supplement learning tool for understanding westward expansion, the industrial revolution, and the early use of railroads. Along these lines, having players earn additional points for identifying the railroad companies that historically provided service between destination cities will bring this period in history home.

While Ticket to Ride is obviously well suited to variations that focus on developing an understanding of geography and history, the game also lends itself to variations that focus on math, strategy and team building skills. Based on Ticket to Ride's use of a number line for

scoring, a simple and fun variation is to require players to move back two spaces if their score lands them on a number ending in zero or five. This small variation not only requires players to perform mental math calculations for each potential move, but also requires strategic thinking and reasoning to evaluate the most beneficial move.

As an added modification, identify certain train spaces that will act as multipliers for the route score. For example, a space on the three-train route between Las Vegas and Los Angeles can be designated as a "3×" score, thereby tripling the score for that route (think Scrabble). Not only does this add in multiplication skills to the game play, but it also adds a wrinkle to the strategy and planning for each player.

Finally, Ticket to Ride is a perfect game to be played in teams. However, to maximize the fun and benefit of team play create special destination cards that only have one of the destinations listed. Each teammate only gets a card(s) with one of the destination cities, which they cannot share with their teammate. Teams then work to complete their destination routes by observing their teammates moves and deducing the destination cities in their hand; while not making it so obvious that the other team will move to block those routes. As a side, team play using these rules is equally as fun for adults, although I can attest to the fact that you may not want your spouse as your teammate!

OBSERVATIONS ON TICKET TO RIDE

- Ticket to Ride—Nordic Countries was initially only to be sold in those countries.
- Zug um Zug: Deutschland is only sold in German and Austria.
- Rails & Sails edition of Ticket to Ride has a Great Lakes map that uses ships, as well as a world map that uses ships and trains.
- A card game version of Ticket to Ride was released in 2008.
- The iPad version of Ticket to Ride was named the 2011 Digital Game of the Year by the Danish Guldbrikken Awards.

- A secondary market exists for customized boards and destination tickets for Ticket to Ride.
- The fictional inspiration for Ticket to Ride is the world traveler, Phileas Fogg.
- A statistical analysis of the most “valuable” cities of the original Ticket to Ride game identified Los Angeles, Miami, New York, and Vancouver.
- Ticket to Ride has won over thirty-five international game awards.
- The Mystery Train add-on was introduced as a give-away in the December 2004 *Game Trade Magazine*. This also marked the introduction of character cards.
- The song “Ticket to Ride” was the seventh consecutive number one hit by the Beatles in the United Kingdom in April 1965.
- Some of the game components for Ticket to Ride are manufactured in Lafayette, Indiana.
- Alvin the Alien and Dexter the Dinosaur are part of the expansion pack for Ticket to Ride that combines trains, aliens, and dinosaurs.
- A map design contest for Ticket to Ride—Legendary Asia was won by Francois Valentyne. He won \$10,000.
- Electronic versions of Ticket to Ride unlock facts and pictures of cities once they are connected.
- The Ticket to Ride World Championship was held in Paris, France. The grand prize was a trip for two to Asia with a rail journey on the Orient Express from Bangkok to Singapore. (Hopefully no murder occurred.)

The measure of a good game is how much it gets played and used. We are on our third Ticket to Ride game. Our first game was used by our son as a part of his fifth-grade project on the Industrial Revolution in the United States. From this we learned that train pieces glued to a board with Gorilla Glue do not come off; ever.

Our second game was our family’s game of choice during a beach trip to Hilton Head Island, South Carolina. It was during this trip that we learned, unlike the waterproof UNO cards, the Ticket to Ride board and cards were not waterproof.

Our current version is still in one piece; so far.